

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)

Sem - I University Examination, December- 2012

M.Tech. (CAD/CAM)

ME – 703.01 Advanced dynamics of Machine

Date: 21.12.2012, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

Q1 (a) Perform position analysis of four-bar linkage in Fig.1 by finding θ_3 and θ_4 for the two positions shown. 07

Q.2 (a) In the slider crank mechanism ,the line of stroke of slider P is off-set a perpendicular distance of 5cm, from the crank centre C. Connecting rod PB and crank BC are respectively 75 cm and 20 cm long. Crank BC rotates clockwise at a uniform speed of 200 r.p.m. Find analytically linear velocity of slider and angular velocity of connecting rod. Also find out acceleration of slider and angular acceleration of PB. Take angle DCB= 135° , the datum line DC being parallel to the line of stroke. 07

(b) A four link mechanism is subjected to following external forces as shown in Fig.2. 07

Link	Length	Force	Magnitude
AB	500mm	F2	80N
BC	600mm	F3	144N
CD	560mm	F4	60N
AD	1000mm	-	Fixed link

Find the input torque T_2 required on crank for static equilibrium by using Virtual work method.

OR

Q.2 (a) Fig.3 shows an quick return mechanism. The driving crank OA is 60 mm long and rotates at a uniform speed of 200 r.p.m. in c.w. direction. For the position shown Find velocity of ram R and angular velocity of link BD. 07

(b) Fig.4 Shows a Mechanism ABCD, EFG are the midpoints of the links as shown in fig. Determine the perpendicular force F_2 (Magnitude and Direction) to keep the mechanism in static equilibrium. AB = 30 mm, BC = 90 mm, CD = 60 mm, AD = 40 mm. 07

Q.3 (a) The crank pin circle radius of a horizontal engine is 300mm. The mass of the reciprocating parts is 250kg. When the crank has traveled 60° from I.D.C., the difference between the driving and the back pressures is 0.35N/mm². The connecting 08

rod length between centers is 1.2 m and the cylinder bore is 0.5m. If the engine runs at 250r.p.m. and if the effect of piston rod diameter is neglected, calculate: 1. Pressure on slide bars, 2. thrust in the connecting rod, 3. tangential force on the crank-pin, and 4. turning moment on the crank shaft.

- (b) How will you determine jump speed for an eccentric cam operating a flat faced follower? 06

OR

- Q.3 (a) The following data refer to a steam Engine Determine the magnitude and direction of the torque exerted on the crankshaft when the crank turned through 30° from I.D.C. Diameter of Piston=240mm; stroke=600mm, length of connecting rod=1.5m; mass of reciprocating parts=300kg; mass of connecting rod = 250 kg; speed = 125 r.p.m.; centre of gravity of connecting rod from crank pin = 500mm; radius of gyration of the connecting rod about an axis through the centre of gravity = 650mm. 08
- (b) In 3-4-5 Polynomial cam follower rise 20mm in 120° of cam rotation. The cam shaft rotates at 300 rpm. Find Displacement, Velocity, Acceleration and Jerk when shaft has turned through 35° . 06

SECTION- II

- Q.4 (a) Explain the mobility of spatial mechanism 07

- Q.5 (a) Design and draw a slider crank mechanism in which displacement of slider S is given by following relation : 10

$$S_j = \frac{66825 + 630\theta_j - 7\theta_j^2}{8100}$$

Where θ_j = Crank angle of rotation in the range, $45^\circ \leq \theta \leq 135^\circ$. Use three point Chebyshev spacing.

- (b) Explain Path Generation Synthesis 04

OR

- Q.5 (a) Synthesis a slider crank mechanism for the following position of slider and crank. 08

$$S = 0.9215m, \theta = 30^\circ$$

$$\dot{S} = -2.23m/sec, \omega_2 = 20rad/sec$$

$$\ddot{S} = -83.84m/sec^2, \alpha_2 = 10rad/sec^2$$

Draw the mechanism by using Freudenstien's equation.

- (b) Explain Coupler curve synthesis for 4-bar mechanism. 06

- Q.6 (a) Explain active and passive balancing. Determine the possible firing order for four stroke Six- cylinder in line engine. 06

- (b) Investigate the state of primary and secondary balancing for reciprocating masses of a two stroke cycle six cylinder diesel engine. The firing order of a is I-V-III-VI-II-IV. 08

OR

- Q.6 (a) An engine having five cylinders in line, works on a four stroke cycle. The distance between centre lines is 375mm and the reciprocating mass for each cylinder is 15.875kg, the crank radius is 114.3 mm and the connecting rod length is 450 mm. 08

The engine runs at 600 rpm. Examine the engine for balance of primary and secondary forces and couples. Determine the maximum values of these and the position of the central crank at these maximum values occur. (By Analytical method).

- (b) A three cylinder radial engine has cylinders located 120° from neighboring cylinder. 06
 Reciprocating mass of each cylinder is 1.2kg. Length of crank is 75mm and each connecting rod is 250mm long. Find out maximum primary and secondary unbalance forces if the engine runs at 2500 r.p.m. (By Analytical method-Direct & Reverse crank method).

